Supplementary material for

Worked systems for viability under incomplete observation: exact belief kernels across observation structures, protocols and decentralisation, a monitoring-design census, and certified margins

Amin Abaee

Independent Researcher

ORCID: 0000-0002-0019-1842 amin_abaee@ut.ac.ir

October 15, 2026

This supplement collects the tables behind the counts of the main text and the material removed from it: the master table of the 36 pairs (S1), the whole-universe census with the register conventions that are not used in the main text (S2), the stationary-policy enumeration (S3), the decentralised law tables (S4), the monitoring-design census (S5), the four-state target census (S6), the timing grids (S7), the scaling of the census to larger caps (S8), the continuous-fibre tables (S9), the verifier output and the Lean declarations (S10), and a table of symbols (S11). Theorem, section and equation numbers of the form "Theorem " refer to the main text. Every table is generated from `results.json`.

S1 The master table of the 36 pairs

The earlier editions of the audit worked on the 36 unordered pairs of safe states; Table S1.1 retains this population and extends it with the private-register and optimistic-ledger conventions and the decentralised kernel. The eight pairs through the corner 11 are dead in every class because the corner has no safe instrument (Lemma 2.5(c)). Among the 28 corner-free pairs the dead ones are: institutional (ℓ) 12|21, 12|31, 13|21, 13|31; aggregate 12|21, 13|31; codex (ℓ) 12|21, 12|31, 13|21; full none. The mechanisms are as follows.

- 12|21 under o_Σ: both states have aggregate 3, hence lie in one fibre, and $R(12) \cap R(21) = \{1\} \cap \{2\} = \emptyset$; no first instrument exists. Under the codex (ℓ): the two singleton fibres issue 1 and 2; the successor set is $\{21, 12\}$ and must survive the global alternation from both phases, but $21 \notin \Phi_1$.
- 12|31 and 13|21 under the shared ledger (both observations): the first step issues 1 for the state in $\{12, 13\}$ and 2 for the other; the successor sets $\{21, 22\}$ and $\{22, 12\}$ must lie in $\Phi_1 \cap \Phi_2$, which they do not. Under the free protocol each state keeps its own thread and both pairs are viable; under the private register they are viable as well (the threads never share a fibre with opposite phase).
- 13|31 under o_Σ (both protocols) and under the private codex: both states have aggregate 4 and share a fibre with $R(13) \cap R(31) = \emptyset$ under o_Σ; under o_{id} with the private register the two threads land on 22 after one step with opposite phases, which is a coherence violation.

Under the shared ledger at full observation the pair is viable: the mixed first step $(1, 2)$ sends both states to $22 \in \Phi_1 \cap \Phi_2$.

Table S1.1: Master table: the 36 pairs of safe states and their membership (•) in the kernels. Columns: institutional, aggregate, codex and full under the shared ledger; institutional and codex under the private register (π); codex under the optimistic ledger (ℓ^+); decentralised unanimity. Pairs through the corner 11 are in no kernel.

pair	inst	agg	codex	full	inst$^{\pi}$	codex$^{\pi}$	codex$^{\ell^+}$	dec
11|12	·	·	·	·	·	·	·	·
11|13	·	·	·	·	·	·	·	·
11|21	·	·	·	·	·	·	·	·
11|22	·	·	·	·	·	·	·	·
11|23	·	·	·	·	·	·	·	·
11|31	·	·	·	·	·	·	·	·
11|32	·	·	·	·	·	·	·	·
11|33	·	·	·	·	·	·	·	·
12|13	•	•	•	•	•	•	•	·
12|21	·	·	·	•	·	•	·	•
12|22	•	•	•	•	•	•	•	·
12|23	•	•	•	•	•	•	•	•
12|31	·	•	·	•	•	•	•	·
12|32	•	•	•	•	•	•	•	•
12|33	•	•	•	•	•	•	•	·
13|21	·	•	·	•	•	•	•	·
13|22	•	•	•	•	•	•	•	•
13|23	•	•	•	•	•	•	•	·
13|31	·	·	•	•	·	·	•	•
13|32	•	•	•	•	•	•	•	·
13|33	•	•	•	•	•	•	•	•
21|22	•	•	•	•	•	•	•	·
21|23	•	•	•	•	•	•	•	•
21|31	•	•	•	•	•	•	•	·
21|32	•	•	•	•	•	•	•	•
21|33	•	•	•	•	•	•	•	·
22|23	•	•	•	•	•	•	•	·
22|31	•	•	•	•	•	•	•	•
22|32	•	•	•	•	•	•	•	·
22|33	•	•	•	•	•	•	•	•
23|31	•	•	•	•	•	•	•	·
23|32	•	•	•	•	•	•	•	•
23|33	•	•	•	•	•	•	•	·
31|32	•	•	•	•	•	•	•	·
31|33	•	•	•	•	•	•	•	•
32|33	•	•	•	•	•	•	•	·

S2 The whole-universe census and the register conventions

Table S2.2 gives the viable beliefs by size and the stabilisation index of the horizon family on the whole node set; Table S2.3 lists the inclusion-maximal viable beliefs. All kernels are closed under

sub-beliefs (checked setwise), so each is the down-closure of its maximal members.

Table S2.2: Viable beliefs by size for each class and convention (columns 1–8 are belief sizes), with the total and the number h^* of applications of Φ after which the horizon family on the whole node set is stationary ($\mathcal{K} = \mathcal{W}_{h^*}$).

class	1	2	3	4	5	6	7	8	total	h^*
full	8	28	56	70	56	28	8	1	255	1
codex (ℓ)	8	25	39	32	13	2	0	0	119	2
aggregate	8	26	44	41	20	4	0	0	143	1
institutional (ℓ)	8	24	36	29	12	2	0	0	111	2
codex (π)	8	27	50	55	36	13	2	0	191	2
institutional (π)	8	26	44	41	20	4	0	0	143	1
codex (ℓ^+)	8	27	50	55	36	13	2	0	191	2
institutional (ℓ^+)	8	26	42	35	14	2	0	0	127	2
decentralised	8	12	8	2	0	0	0	0	30	—

Table S2.3: Inclusion-maximal viable beliefs (the antichains that generate the kernels by sub-belief closure).

class	maximal beliefs
full	12|13|21|22|23|31|32|33
codex (ℓ)	13|23|31|32|33, 12|13|22|23|32|33, 21|22|23|31|32|33
aggregate	12|13|22|23|32|33, 12|22|23|31|32|33, 13|21|22|23|32|33, 21|22|23|31|32|33
institutional (ℓ)	12|13|22|23|32|33, 21|22|23|31|32|33
codex (π)	12|13|21|22|23|32|33, 12|21|22|23|31|32|33
institutional (π)	12|13|22|23|32|33, 12|22|23|31|32|33, 13|21|22|23|32|33, 21|22|23|31|32|33
codex (ℓ^+)	12|13|22|23|31|32|33, 13|21|22|23|31|32|33
institutional (ℓ^+)	12|23|31|32|33, 13|21|23|32|33, 12|13|22|23|32|33, 21|22|23|31|32|33
blind	12|13|22|23|32|33, 21|22|23|31|32|33
decentralised	12|21|23|32, 13|22|31|33

The join of the middle classes. On all beliefs, $K_{\text{full}} \setminus (K_{\text{agg}} \cup K_{\text{cod}})$ has 104 members: the 64 beliefs containing $\{12, 21\}$ and the 56 beliefs containing $\{13, 31\}$ together with at least one of $12, 21, 22$, with 16 beliefs in both groups; its minimal members are 12\|21, 12\|13\|31, 13\|21\|31 and 13\|22\|31. The eight beliefs containing $\{13, 31\}$ that are codex-viable are $\{13, 31\} \cup S$ with $S \subseteq \{23, 32, 33\}$. $K_{\text{agg}} \setminus K_{\text{cod}}$ has 32 members with minimal members 12\|31 and 13\|21; $K_{\text{cod}} \setminus K_{\text{agg}}$ has 8 members with minimal member 13\|31.

Conventions not used in the main text. Two further conventions were reported in earlier editions: the ledger records the instrument of the *first* or of the *last* fibre in a fixed order of the fibres (by their smallest state). Both give $(25, 26, 26, 28)$ on the pair population, they disagree with each other as sets, and neither coincides with the shared-ledger kernels $(24, 26, 25, 28)$; this is why the shared ledger quantifies over all issued instruments instead of a fixed thread. The optimistic ledger gives $(26, 26, 27, 28)$ on pairs and $(127, 143, 191, 255)$ on beliefs; its codex kernel has the same size as the private-register codex kernel (191) but is a different set (its maximal beliefs are 12\|13\|22\|23\|31\|32\|33 and 13\|21\|22\|23\|31\|32\|33 against 12\|13\|21\|22\|23\|32\|33 and 12\|21\|22\|23\|31\|32\|33). Admitting the idle instrument 0 as a third option leaves all four shared-ledger kernels unchanged on all 511 beliefs of safe states: it is dominated.

Closed forms. The closed forms of Theorem 4.3 were implemented without the game engine (alternating trajectories and, for the private register, the coherence check over 24 steps) and agree with the engine on all 511 beliefs of safe states for the institutional and codex classes under both conventions. The blind free kernel is $(2^{\Phi_1} \cup 2^{\Phi_2}) \setminus \{\emptyset\}$ with 111 members and equals the shared-ledger institutional kernel setwise.

S3 Stationary policies

A deterministic stationary policy fixes the closed loop, so a belief is sustained by it exactly when each of its states is; Table S3.4 reports the unions over policy classes of the sustained beliefs (Proposition 4.5). The four largest sets sustained by single alternation policies at aggregate observation are the four maximal beliefs of the aggregate kernel, 12|13|22|23|32|33, 12|22|23|31|32|33, 13|21|22|23|32|33 and 21|22|23|31|32|33; they correspond to the first instruments $(1, 1, \cdot, \cdot)$, $(1, 2, \cdot, \cdot)$, $(2, 1, \cdot, \cdot)$ and $(2, 2, \cdot, \cdot)$ on the aggregate cells 3 and 4 (cells 5 and 6 admit either). The memoryless aggregate policies sustain no state because the aggregate is non-increasing along every safe trajectory, so a memoryless aggregate policy eventually repeats one instrument, and a repeated instrument drives the other stock to 0 within three steps. Enumerating the 2^{27} full-observation policies with memory is unnecessary: the 512 memoryless ones already realise the full kernel.

Table S3.4: Stationary-policy unions: number of deterministic policies, number of beliefs (of 255) sustained by at least one of them, and the largest number of states sustained by a single policy.

observation	policy class	policies	beliefs	max states
o_Σ	memoryless, $o_\Sigma(V) \to A$	32	0	0
o_Σ	one period of instrument memory	32768	143	6
o_Σ	alternation (first instrument per cell)	32	143	6
o_{id}	memoryless, $V \to A$	512	255	8

S4 Decentralised unanimity: law tables

Table S4.5 gives, for each safe state, the number of law pairs (of 256) that sustain it as a singleton; by Theorem 5.2 a belief is sustained if and only if these sets intersect. The 30 sustained beliefs are the nonempty subsets of $\{12, 21, 23, 32\}$ and of $\{13, 22, 31, 33\}$ (two 4-sets: $2 \cdot 15 = 30$), so the sustained pairs are the 12 pairs inside the two maximal beliefs and the family $\{\mathcal{S}(x)\}$ has the Helly property on this instance. Table S4.6 lists the 16 law pairs sustaining 12|21 and Table S4.7 the authority census. The 256 law pairs fall into 64 behavioural classes (votes at $z = 0$ are never consulted on safe orbits); the 16 sustaining pairs of 12|21 are 4 behavioural classes, each extended by the 4 unconsulted vote pairs. Fixing $l_1(1) = 1$ leaves all 16 sustaining pairs of 12|21, fixing $l_2(1) = 1$ leaves none.

Table S4.5: Number of law pairs (of 256) sustaining each safe state as a singleton.

state	$\lvert\mathcal{S}(x)\rvert$
11	0
12	16
13	20
21	16
22	32
23	24
31	20
32	24
33	16

Table S4.6: The 16 law pairs sustaining 12|21: the product $L_1 \times L_2$ (rows L_1, columns L_2; laws written as the votes at $z = 0, 1, 2, 3$).

$l_1 \backslash l_2$	1211	1212	2211	2212
1121	(1121,1211)	(1121,1212)	(1121,2211)	(1121,2212)
1122	(1122,1211)	(1122,1212)	(1122,2211)	(1122,2212)
2121	(2121,1211)	(2121,1212)	(2121,2211)	(2121,2212)
2122	(2122,1211)	(2122,1212)	(2122,2211)	(2122,2212)

Table S4.7: Authority census: the pair kernel when one vote at the floor is fixed, and when a whole law is fixed to a constant.

restriction	pairs	sustained pairs
$l_1(1) = 1$ (owner)	12	12|21, 12|23, 12|32, 13|22, 13|31, 13|33, 21|23, 21|32, 22|31, 22|33, 23|32, 31|33
$l_1(1) = 2$ (against)	4	22|31, 22|33, 23|32, 31|33
$l_2(1) = 1$ (against)	4	13|22, 13|33, 22|33, 23|32
$l_2(1) = 2$ (owner)	12	12|21, 12|23, 12|32, 13|22, 13|31, 13|33, 21|23, 21|32, 22|31, 22|33, 23|32, 31|33
$l_1 \equiv 1$	0	—
$l_1 \equiv 2$	0	—
$l_2 \equiv 1$	0	—
$l_2 \equiv 2$	0	—

S5 The monitoring-design census

All $B_8 = 4{,}140$ set partitions of the eight viable states were enumerated and the free-protocol kernel of each computed by the bitmask implementation of the belief game (greatest fixed point on the 255 beliefs). Table S5.8 gives the distribution of kernel sizes, Table S5.9 the sixteen coarsest adequate designs. The count of adequate designs by the closed form (3) is $151 + 372 + 372 + 799 = 1{,}694$: for the independent-set partitions π of the critical four states $\{12, 13, 21, 31\}$ with $|\pi| = 2, 3, 3, 4$ blocks, $\sum_{m=0}^{4} \binom{4}{m} B_m |\pi|^{4-m}$ equals $16 + 32 + 48 + 40 + 15 = 151$ ($|\pi| = 2$), $81 + 108 + 108 + 60 + 15 = 372$ ($|\pi| = 3$) and $256 + 256 + 192 + 80 + 15 = 799$ ($|\pi| = 4$). Dynamic adequacy coincides with one-step adequacy on all 4,140 partitions, as Theorem 6.3 requires. The minimum kernel 111 is attained by exactly the 52 partitions in which one cell contains all four

critical states (the number of partitions of the remaining four states into blocks that may or may not join that cell: $\sum_m \binom{4}{m} B_m = 1 + 4 + 12 + 20 + 15 = 52$), and every such kernel equals the blind kernel. The aggregate partition $\{12, 21\} \,|\, \{13, 22, 31\} \,|\, \{23, 32\} \,|\, \{33\}$ has kernel 143 with four maximal beliefs of size 6 and 26 viable pairs. The horizon family stabilises after 0, 1 or 2 applications of Φ for 1,694, 1,404 and 1,042 partitions respectively. Under the shared-ledger no-repeat protocol the closed form of Theorem 4.3(a) was evaluated for every partition: no partition sustains the whole viable set and the largest kernel is 119 (the codex kernel, attained at the finest partition).

Table S5.8: Distribution of kernel sizes over the 4,140 monitoring partitions (18 sizes, 74 distinct kernels).

kernel size	partitions
111	52
123	20
129	10
131	54
133	17
135	204
137	12
139	24
143	217
155	102
159	348
167	210
173	16
175	172
179	39
183	54
191	895
255	1694

Table S5.9: The 16 coarsest (two-cell) dynamically adequate monitoring designs.

design
12|13 | 21|22|23|31|32|33
12|13|22 | 21|23|31|32|33
12|13|23 | 21|22|31|32|33
12|13|22|23 | 21|31|32|33
12|13|22|23|32|33 | 21|31
12|13|23|32|33 | 21|22|31
12|13|22|32|33 | 21|23|31
12|13|32|33 | 21|22|23|31
12|13|32 | 21|22|23|31|33
12|13|22|32 | 21|23|31|33
12|13|23|32 | 21|22|31|33
12|13|22|23|32 | 21|31|33
12|13|22|23|33 | 21|31|32
12|13|23|33 | 21|22|31|32
12|13|22|33 | 21|23|31|32
12|13|33 | 21|22|23|31|32

S6 The four-state target

Table S6.10 lists the $B_4 = 15$ partitions of $\{12, 21, 22, 31\}$ with their one-step adequacy. The incompatibility graph has the two edges $12 - 21$ and $12 - 31$; 5 partitions merge the first edge, 5 the second and 2 both, so $15 - (5 + 5 - 2) = 7$ are adequate; the two adequate two-cell partitions are the two proper 2-colourings of the path $21 - 12 - 31$ with the isolated 22 attached to either side. The maximal common-action sets are $\{12, 22\}$ and $\{21, 22, 31\}$. The Helly numbers of the families of all nonempty subsets of a two- and a three-element instrument set are 2 and 3 (Theorem 6.2(b)), computed by enumerating inclusion-minimal subfamilies with empty intersection.

Table S6.10: The 15 partitions of the four-state target $\{12, 21, 22, 31\}$ and their one-step adequacy (•: every cell has a common safe instrument).

partition	cells	adequate
12 | 21 | 22 | 31	4	•
12 | 21 | 22|31	3	•
12 | 21|22 | 31	3	•
12 | 21|22|31	2	•
12 | 21|31 | 22	3	•
12|21 | 22 | 31	3	·
12|21 | 22|31	2	·
12|21|22 | 31	2	·
12|21|22|31	1	·
12|21|31 | 22	2	·
12|22 | 21 | 31	3	•
12|22 | 21|31	2	•
12|22|31 | 21	2	·
12|31 | 21 | 22	3	·
12|31 | 21|22	2	·

S7 The timing grids

Table S7.11 is the 93-cell grid of Corollary 7.4: 33 viable cells under free switching with unit damage ($D_T = T$) and 43 under alternation with damages $(1, \frac{1}{2})$ ($D_1, D_2, D_3 = 1, \frac{3}{2}, \frac{5}{2}$; 21, 16 and 6 viable cells for $T = 1, 2, 3$). The strict form $z_0 - 1 > T$ of the free identity fails exactly at $(2.0, 1)$, $(3.0, 2)$, $(4.0, 3)$; the strict form against the integer count $\lfloor z_0 - 1 \rfloor > T$ fails on 21 cells, one in each row with $z_0 \geq 2$. On the three-regime graph without self-loops and damages $(1, \frac{1}{2}, \frac{1}{4})$ the max-plus recursion and path enumeration agree for $T \leq 6$ with $D_T = 1, \frac{3}{2}, \frac{5}{2}, 3, 4, \frac{9}{2}$.

Table S7.11: The 93-cell timing grid: for each initial level z_0 (slack $z_0 - 1$, integer count $\lfloor z_0 - 1 \rfloor$) and review period T, viability under free switching with unit damage (f) and under alternation with damages $(1, \frac{1}{2})$ (a).

z_0	$\lfloor z_0 - 1 \rfloor$	T=1 f	a	T=2 f	a	T=3 f	a
1.0	0	·	·	·	·	·	·
1.1	0	·	·	·	·	·	·
1.2	0	·	·	·	·	·	·
1.3	0	·	·	·	·	·	·
1.4	0	·	·	·	·	·	·
1.5	0	·	·	·	·	·	·
1.6	0	·	·	·	·	·	·
1.7	0	·	·	·	·	·	·
1.8	0	·	·	·	·	·	·
1.9	0	·	·	·	·	·	·
2.0	1	•	•	·	·	·	·
2.1	1	•	•	·	·	·	·
2.2	1	•	•	·	·	·	·
2.3	1	•	•	·	·	·	·
2.4	1	•	•	·	·	·	·
2.5	1	•	•	·	•	·	·
2.6	1	•	•	·	•	·	·
2.7	1	•	•	·	•	·	·
2.8	1	•	•	·	•	·	·
2.9	1	•	•	·	•	·	·
3.0	2	•	•	•	•	·	·
3.1	2	•	•	•	•	·	·
3.2	2	•	•	•	•	·	·
3.3	2	•	•	•	•	·	·
3.4	2	•	•	•	•	·	·
3.5	2	•	•	•	•	·	•
3.6	2	•	•	•	•	·	•
3.7	2	•	•	•	•	·	•
3.8	2	•	•	•	•	·	•
3.9	2	•	•	•	•	·	•
4.0	3	•	•	•	•	•	•

S8 Scaling to larger caps

The worked system with cap c has states $\{0, \ldots, c\}^2$, the same instruments and the same floor. Table S8.12 gives the pair census for $c = 3, 4, 5$: in every case the full kernel is the set of corner-free pairs, the institutional kernel under the private register equals the aggregate kernel, and all inclusions of Corollary 3.2 are strict, with the aggregate and codex kernels incomparable. The

computation uses the generic game engine on the reachable node graph of the pair population.

Table S8.12: Scaling of the pair census to larger caps: safe states, pairs of safe states, pair kernels of the four classes under the shared ledger, the institutional and codex pair kernels under the private register, and the number of corner-free pairs (which equals the full kernel in every case).

cap	safe	pairs	inst	agg	codex	full	inst^{π}	codex^{π}	corner-free
3	9	36	24	26	25	28	26	27	28
4	16	120	98	102	100	105	102	104	105
5	25	300	266	272	269	276	272	275	276

S9 The continuous fibre

Table S9.13 tabulates the benchmark fibre of Proposition 8.1 and Table S9.14 the drift audit of Proposition 8.4. The duality values of Proposition 8.2 are computed on the rational grids $u \in \{0, \frac{1}{100}, \dots, 1\}$ and $\lambda \in \{0, \frac{1}{100}, \dots, 1\}$: primal and dual $-\frac{1}{10}$ at $u = \frac{1}{2}$ and $\lambda = (\frac{1}{2}, \frac{1}{2})$ for the two-floor instance; primal -1 and dual 0 on the nonconvex control set $\{0, 1\}$ with payoff rows $(-1, +1)$ and $(+1, -1)$; primal and dual $+\frac{1}{10}$ at $u = \frac{3}{10}$ for the feasible contrast. The Helly-tight triple $\{u_1 \geq 1, u_2 \geq 1, u_1 + u_2 \leq 1\}$ is pairwise feasible on the half-integer grid and jointly infeasible. The two floors $u - \frac{1}{2} \geq 0$ and $-u - \frac{1}{2} \geq 0$ on $[-1, 1]$ are each satisfiable and jointly unsatisfiable, with $\max_u \min_j f_j(u) = -\frac{1}{2}$.

Table S9.13: The benchmark fibre: caps, cap sum and status against the demand $u_1 + u_2 \geq 2$; the margin is the dual-weighted deficiency $\mu(Y) = (Y - 27/5)/10$ at $\lambda = (\frac{1}{2}, \frac{1}{2})$.

Y	$c_1(Y)$	$c_2(Y)$	$c_1 + c_2$	status
4	$\frac{13}{10}$	$\frac{49}{50}$	$\frac{57}{25}$	feasible
$\frac{9}{2}$	$\frac{5}{4}$	$\frac{93}{100}$	$\frac{109}{50}$	feasible
5	$\frac{6}{5}$	$\frac{22}{25}$	$\frac{52}{25}$	feasible
$\frac{27}{5}$	$\frac{29}{25}$	$\frac{21}{25}$	2	feasible
$\frac{11}{2}$	$\frac{23}{20}$	$\frac{83}{100}$	$\frac{99}{50}$	obstructed, margin $\frac{1}{100}$
6	$\frac{11}{10}$	$\frac{39}{50}$	$\frac{47}{25}$	obstructed, margin $\frac{3}{50}$
7	1	$\frac{17}{25}$	$\frac{42}{25}$	obstructed, margin $\frac{4}{25}$

Table S9.14: Certainty-equivalent drift under $g(s) = s^2$ with lead $\frac{1}{10}$: naive drift $s/5 + 1/100$ and corrected drift.

s	naive drift	corrected drift
1	$\frac{21}{100}$	0
$\frac{5}{4}$	$\frac{13}{50}$	0
$\frac{3}{2}$	$\frac{31}{100}$	0
$\frac{7}{4}$	$\frac{9}{25}$	0
2	$\frac{41}{100}$	0

S10 Verification record

Running the verifier. `python3 paper11c_worked_systems_audit_v5_verification.py` (in the directory containing `work/`) recomputes every object from `work/ws_core.py`, evaluates

the 64 checks of Table S10.15, compares the numbers quoted in the compiled manuscript and supplement with the recomputed values when the PDFs are present, and exits with status 0 only if everything passes. The design census (all 4,140 partitions, free protocol and shared ledger) accounts for most of the running time.

Table S10.15: The named checks of the verifier and their status in the shipped run.

status	check
PASS	S1 state grid 16 states, 9 safe, corner 11 has no safe instrument, eight other safe states have R as in eq. (R)
PASS	S2 aggregate z1+z2 is non-increasing under instruments 1 and 2 on all safe states (and falls only when the raised stock is at its cap)
PASS	S3 36 pairs of safe states, 28 corner-free; 511 nonempty beliefs of safe states, 255 of viable states
PASS	K1 shared-ledger pair kernels (inst, agg, codex, full) = (24, 26, 25, 28)
PASS	K2 shared-ledger belief kernels on the 511 beliefs = (111, 143, 119, 255), all contained in the corner-free beliefs
PASS	K3 full kernel = all nonempty subsets of the eight viable states; on pairs = the 28 corner-free pairs
PASS	K4 monotonicity inclusions inst <= agg <= full and inst <= codex <= full hold on all beliefs and all are strict
PASS	K5 aggregate and codex kernels incomparable; witnesses 12|31, 13|21 in agg \{} codex and 13|31 in codex \{} agg
PASS	K6 meet of the middle classes is the institutional kernel on pairs and on all beliefs
PASS	K7 join of the middle classes misses the full kernel by exactly 12|21 on pairs and by 104 beliefs (64 containing 12|21, the rest containing 13|31 plus one of 12, 21, 22); minimal members 12|21, 12|13|31, 13|21|31, 13|22|31
PASS	K8 every kernel (all classes, all conventions) is closed under nonempty sub-beliefs
PASS	K9 maximal beliefs: inst 2x6, agg 4x6, codex 2x6+1x5, full 1x8 (shared ledger)
PASS	K10 horizon family on the whole node set stabilises after one application for the free classes and two for the ledger classes
PASS	K11 pair kernels are the size-2 members of the belief kernels (consistency of the two populations)
PASS	K12 admitting the idle instrument 0 leaves all four shared-ledger kernels unchanged
PASS	A1 alternation-safe sets Phi_1 = {12,13,22,23,32,33}, Phi_2 = {21,22,23,31,32,33}; union = viable states, intersection = {22,23,32,33}
PASS	A2 every alternating trajectory in Phi_a is 2-periodic after at most one step
PASS	A3 shared-ledger closed form reproduces the institutional and codex kernels on all 511 beliefs
PASS	A4 private-register closed form (phase coherence) reproduces the private institutional and codex kernels on all 511 beliefs
PASS	A5 blind kernel = (2^Phi_1 U 2^Phi_2) minus empty, 111 beliefs, equal to the shared-ledger institutional kernel
PASS	A6 codex minus institutional (shared ledger) = the 8 beliefs {13,31} U S, S subset of {23,32,33}

status	check
PASS	A7 private-register kernels (inst, agg, codex, full) = (143, 143, 191, 255) on beliefs and (26, 26, 27, 28) on pairs; private institutional = aggregate setwise
PASS	A8 shared ledger and private register incomparable at full observation: 13|31 ledger-viable and private-dead, 12|21 the reverse
PASS	A9 optimistic-ledger kernels = (127, 143, 191, 255) on beliefs, (26, 26, 27, 28) on pairs; contain the shared-ledger kernels; optimistic codex differs setwise from private codex
PASS	A10 fixed-thread conventions (first / last fibre) give (25, 26, 26, 28) on pairs and disagree with each other setwise
PASS	P1 32 memoryless aggregate policies sustain no state
PASS	P2 32 alternation policies realise exactly the aggregate kernel (143 beliefs; four sustained sets of size 6 = the maximal beliefs)
PASS	P3 all 32768 one-period-memory aggregate policies realise the same set as the 32 alternation policies
PASS	P4 512 memoryless full-observation policies realise the full kernel; the displayed policy sustains all eight viable states
PASS	D1 16 laws, 256 law pairs, 64 behavioural classes
PASS	D2 statewise factoring: the factored kernel equals the direct orbit computation on all 511 beliefs
PASS	D3 decentralised kernel: 12 pairs, 30 beliefs (8+12+8+2), maximal beliefs 12|21|23|32 and 13|22|31|33; 16 of 28 coordinated-viable pairs lost; contains 12|21, misses 12|13
PASS	D4 sustained pairs are exactly the pairs inside the two maximal beliefs (Helly property of the sustainer family on this instance)
PASS	D5 the 16 law pairs sustaining 12|21 form the product L1 x L2 (4 behavioural classes x 4 unconsulted votes)
PASS	D6 authority census: l1(1)=1 and l2(1)=2 leave 12 pairs; l2(1)=1 leaves {13|22,13|33,22|33,23|32}; l1(1)=2 its swap image; constant laws leave 0
PASS	D7 fixing l1(1)=1 leaves 16 sustaining pairs for 12|21, fixing l2(1)=1 leaves 0
PASS	M1 incompatibility graph on the eight viable states = K_{2,2} between {12,13} and {21,31} (4 edges)
PASS	M2 four-state target: 15 partitions, 7 adequate = 15-(5+5-2); edges 12-21, 12-31; two adequate two-cell partitions; maximal common-action sets 12|22 and 21|22|31
PASS	M3 Helly numbers of the two- and three-instrument universes are 2 and 3
PASS	M4 design census: 4140 partitions, 1694 dynamically adequate, dynamic adequacy = one-step adequacy on every partition
PASS	M5 closed-form count 151 + 2*372 + 799 = 1694 agrees with the census
PASS	M6 16 coarsest (two-cell) adequate designs; 74 distinct kernels; 18 kernel sizes; minimum 111 attained by exactly the 52 partitions merging the four critical states
PASS	M7 maximal-belief sizes 6/7/8 for 927/1519/1694 partitions; pair kernels 24/26/27/28 for 52/906/1488/1694; aggregate partition: kernel 143, 4 maximal beliefs, 26 pairs, not one-step adequate
PASS	M8 under the shared ledger no partition is adequate and the largest kernel over all partitions is 119
PASS	R1 frozen regimes: pinning identity, all 9 states viable, 36 pairs each

status	check
PASS	R2 mode-blind kernel = the eight corner-free states (28 pairs), equal to the full-information kernel; corner lost
PASS	R3 mode-blind horizon family stabilises after one application (W_1 = W_2 = W_3)
PASS	T1 timing grid: 93 cells, 33 viable under free switching, 43 under alternation (21, 16, 6)
PASS	T2 strict form fails exactly at (2,1), (3,2), (4,3); floored strict form fails on 21 cells
PASS	T3 alternation worst damage 1, 3/2, 5/2, 3, 4, 9/2 = ceil(T/2) + floor(T/2)/2; max-plus recursion equals path enumeration on the 3-regime graph
PASS	C1 benchmark crossover Y* = 27/5; margin law (Y-27/5)/10; Y=6 caps 11/10, 39/50, sum 47/25, margin 3/50; Y=5 witness (6/5, 4/5)
PASS	C2 radius: per-cap increase below 3/50 leaves the Y=6 obstruction, 3/50 removes it
PASS	C3 two-floor duality: primal = dual = -1/10 at u = 1/2, lambda = (1/2, 1/2)
PASS	C4 nonconvex gap: primal -1 against dual 0; feasible contrast +1/10 on both sides at u = 3/10
PASS	C5 Helly-tight triple pairwise feasible, jointly infeasible
PASS	C6 drift: naive s/5 + 1/100 (21/100 ... 41/100), corrected 0, sign law
PASS	C7 floors: each satisfiable, no common instrument, value -1/2
PASS	F1 cap 4: 120 pairs, kernels (98, 102, 100, 105), full = corner-free, private institutional = aggregate
PASS	F2 cap 5: 300 pairs, kernels (266, 272, 269, 276), full = corner-free, private institutional = aggregate
PASS	J1 results.json agrees with the recomputation on the headline counts
PASS	L1 Lean modules present with theorems Wk_mono_obs, master_action_axis, master_memory_axis, master_monotone and no sorry
PASS	X1 main text quotes the recomputed headline numbers (111, 143, 119, 255, 4,140, 1,694, 74 distinct kernels, 27/5, 3/50, -1/10, 12 of the 28, 16 coarsest, 32 alternation policies, 33 and 43, 93-cell)
PASS	X2 main text states the check count (64 named checks) consistently with this script
PASS	X3 supplement quotes the recomputed numbers (104 members, 151+372+372+799, 4 behavioural classes, 52 partitions, (25, 26, 26, 28), (127,143,191,255))

Lean modules. Table S10.16 lists the declarations of the two Lean modules shipped with the article; they import the `Prelude` and `P1_Obstruction` modules of the repository's `Formalizations` development and formalise the three axes of Theorem 3.1 as horizon-indexed inclusions. The verifier checks the presence of the declarations in the shipped sources.

Table S10.16: Declarations of the Lean modules shipped with the article.

module	kind	name
`WS_WorkedSystems.lean`	def	`refineObs`
`WS_WorkedSystems.lean`	theorem	`Wk_mono_obs`
`WS_WorkedSystems.lean`	theorem	`master_action_axis`
`WS_WorkedSystems.lean`	theorem	`master_memory_axis`
`WS_WorkedSystems_v2.lean`	theorem	`master_monotone`
`WS_WorkedSystems_v2.lean`	theorem	`monotone_action_axis`

S11 Symbols

symbol	meaning
X, V	state grid $\{0,\dots,3\}^2$ and safe set $\{z_1 \geq 1, z_2 \geq 1\}$ (nine states)
$A = \{1,2\}$, 0	instruments; idle instrument
F_u	transition of instrument u; $F_u(C)$ image of a set
$R(x)$, $R(C)$	safe instruments of a state, common safe instruments of a set
$\mathsf{z_1z_2}$, $\mathsf{x}\vert\mathsf{y}$	state and belief labels
o, o_{id}, o_{Σ}, o_{b}	observation map; full, aggregate, blind
B_y	fibre of the belief B at observation y
f, nr	free and no-repeat protocols
ℓ, π, ℓ^{+}	shared ledger, private register, optimistic ledger
$\mathcal{N}$, Φ, $\mathcal{W}_h$, $\mathcal{K}$	nodes, operator, horizon family, kernel of the belief game
$K(o,P)$	viable beliefs under observation o and protocol P
$K_{\mathrm{full}}, K_{\mathrm{cod}}, K_{\mathrm{agg}}, K_{\mathrm{inst}}, K_{\mathrm{dec}}$	kernels of the full, codex, aggregate, institutional and decentralised classes
V^*, R^*, G^*	viable singletons, instruments keeping a state in V^*, incompatibility graph on V^*
Φ_a	states surviving the alternating schedule that starts with instrument a
$I(a)$	set of instruments issued by a profile
l_i, T_{l_1,l_2}, $\mathcal{S}(x)$	agency laws, unanimity closed loop, law pairs sustaining x
G_T, $N(G_T)$, B_m	incompatibility graph of a target, number of independent-set partitions, Bell numbers
F_u^r, D_T	regime dynamics; worst cumulative damage over T periods
$c_i(Y)$, Y^*, $\mu(Y)$	caps, crossover and dual margin of the benchmark fibre